

Chern classes

Euler class

Let $\zeta: E \rightarrow B$ be a rank n $\mathbb{R}$ -vector bundle. The presheaf

$$U \mapsto H^*(E|_U, E|_U \otimes \mathbb{Z})$$

is a sheaf. An orientation of ζ is a choice of generator u of $H^*(E|_U, E|_U \otimes \mathbb{Z})$.

We have

$$H^*(E|_U, E|_U \otimes \mathbb{Z}) \xrightarrow{\cong} H^*(E|_U) \xrightarrow{\pi^*} H^*(B)$$

and we denote the image of the orientation by $e(\zeta) = \pi^*(u|_E) \in H^*(B)$, its Euler class.

Example (Euler class of normal bundle): Let $M^n \subset \mathbb{A}^{n+k}$ with normal bundle $\nu: E \rightarrow M$. Taking a tubular neighbourhood, we get

$$u \in H^*(E, E \otimes \mathbb{Z}) \cong H^*(A, A \otimes M).$$

Then $e(\nu)$ is the image of u under the composition

$$H^*(A, A \otimes M) \rightarrow H^*(A) \rightarrow H^*(M).$$

Note that the tangent space of M is the normal bundle of $M \rightarrow M \times M$ via

$$(u, v) \in T_x(M \times M) = T_x M \times T_x M$$

is in $T_x M$ iff $u=v$, and in ν iff $u=-v$, so

$$TM \rightarrow \nu, (x, v) \mapsto (x, v, -v)$$

is an isomorphism.

Chern classes

Let $w: E \rightarrow B$ be a $\mathbb{C}^n$ -vector bundle. Then $e(w_R) \in H^{2n}(B)$. We will inductively define its Chern classes. Consider the l.e.s.

$$\begin{array}{ccccc} \dots & \rightarrow & H^i(E, E \otimes \mathbb{Z}) & \rightarrow & H^i(E) & \xrightarrow{\pi^*} & H^i(B) & \rightarrow & \dots \\ & & \uparrow \cong & & \uparrow \cong & & \uparrow \cong & & \\ & & H^{i-2n}(E) & & H^{i-2n}(B) & & H^{i-2n}(E) & & \\ & & \cong H^{i-2n}(B) & & & & \cong H^{i-2n}(B) & & \end{array}$$

We get the Gysin sequence

$$\dots \rightarrow H^{i-2n}(B) \xrightarrow{\wedge} H^i(B) \xrightarrow{\pi^*} H^i(E \otimes \mathbb{Z}) \rightarrow H^{i-2n+1}(B) \rightarrow \dots$$

In particular, for $j \leq 2n-1$, $H^j(B) \xrightarrow{\pi^*} H^j(E \otimes \mathbb{Z})$.

We define

$$c_n(w) = e(w_R) \in H^{2n}(B)$$

$$c_i(w) = (\pi^*)^{-1} c_i(w_R) \in H^{2n-2i}(B)$$

for $i < n$, where w_R is constructed in

$$\begin{array}{ccc} & \pi^* E & \rightarrow \pi^* E / \mathcal{S} \\ & \downarrow & \swarrow w_R \\ \text{total space} & E \otimes \mathbb{Z} & \\ & \downarrow & \\ & B & \end{array}$$

Prop: Let $\varepsilon: B \times \mathbb{C} \rightarrow B$. Then $c(w \otimes \varepsilon) = c(w)$.

Proof: We have a section

$$\begin{array}{ccc} E & & \\ \downarrow \text{ } \mathcal{S} & & \\ B & & \end{array}$$

We first note that this implies that $e(w \otimes \varepsilon) = c_{n+1}(w \otimes \varepsilon) = 0 = c_{n+1}(w)$:

$$\begin{array}{ccccc} B & \xrightarrow{\varepsilon} & E \otimes \mathbb{Z} & \rightarrow & E & \xrightarrow{\pi^*} & B \\ & & \downarrow \text{ } \mathcal{S} & & \downarrow & & \\ & & E \otimes \mathbb{Z} & & E & & \end{array}$$

For the other Chern classes, we have

$$\begin{array}{ccc} E & \rightarrow & E / \text{ker } \mathcal{S} \rightarrow E' \\ \downarrow w \otimes \varepsilon & & \downarrow w \otimes \varepsilon & \downarrow w \\ B & \xrightarrow{\varepsilon} & E \otimes \mathbb{Z} \xrightarrow{\pi^*} B \end{array}$$

By naturality, $c_i(w \otimes \varepsilon) = \mathcal{S}^* c_i(E / \text{ker } \mathcal{S}) = \mathcal{S}^* \pi^* c_i(w)$.

The Grassmannians $G_n(\mathbb{C}^{n+k})$

First, we calculate the cohomology ring of $\mathbb{P}^k = G_1(\mathbb{C}^{k+1})$. It has a canonical line bundle γ^* . Applying the Gysin sequence, we get

$$\dots \rightarrow H^{i+1}(E \otimes \mathbb{Z}) \rightarrow H^i(\mathbb{P}^k) \xrightarrow{\wedge} H^{i+2}(\mathbb{P}^k) \rightarrow H^{i+2}(E \otimes \mathbb{Z}) \rightarrow \dots$$

Hence, for $0 \leq i \leq 2k-2$, $H^i(\mathbb{P}^k) \xrightarrow{\wedge} H^{i+2}(\mathbb{P}^k)$.

So $H^*(\mathbb{P}^k) \cong \mathbb{Z}[c_1] / c_1^{k+1}$, and $H^*(\mathbb{P}^n) = \mathbb{Z}[c_1]$.

Now, we calculate the cohomology ring of $G_n = G_n(\mathbb{C}^\infty)$. It comes with a taut. bundle γ^* . Apply the Gysin sequence,

$$\dots \rightarrow H^i(G_n) \xrightarrow{\wedge} H^{i+2n}(G_n) \xrightarrow{\pi^*} H^{i+2n}(E \otimes \mathbb{Z}) \rightarrow H^{i+2n}(G_n) \rightarrow \dots$$

We will show that $H^*(E \otimes \mathbb{Z}) = H^*(G_{n-1})$:

Define a map $f: E \otimes \mathbb{Z} \rightarrow G_{n-1}, (X, v) \mapsto X \wedge v^*$. To show that f induces an isomorphism in cohomology, we restrict it to the subbundle of γ^* of n -planes in $\mathbb{C}^n$. If $Y \in G_{n-1}(\mathbb{C}^n)$ then $f|_Y(v) = Y \wedge v^*$ for $v \in Y^\perp \otimes \mathbb{C}^n$, so $f|_Y$ is the projection from W^{n-n+1} to $G_{n-1}(\mathbb{C}^n)$, where W is the orthogonal to Y . It has real rank $2(N-n+1)$, so applying the Gysin sequence, $f|_Y$ induces isomorphisms in cohomology of degrees $\leq 2(N-n)$. Take $N \rightarrow \infty$.

We get the Gysin sequence

$$\dots \rightarrow H^i(G_n) \xrightarrow{\wedge} H^{i+2n}(G_n) \xrightarrow{\pi^*} H^{i+2n}(G_{n-1}) \rightarrow H^{i+2n}(G_n) \rightarrow \dots$$

We now show that $(f^*)^{-1} \pi^*$ maps $c_i(\gamma^*) \mapsto c_i(\gamma^{n-1})$:

$$\begin{array}{ccc} \text{This is clear for } i=n \text{ (by exactness). For } i < n, \\ \pi^* c_i(\gamma^*) \stackrel{\text{def}}{=} c_i(\gamma^* / \text{ker } \mathcal{S}) = f^* c_i(\gamma^{n-1}) \end{array}$$

$$\begin{array}{ccc} \text{since } & \gamma^* / \text{ker } \mathcal{S} & \rightarrow \gamma^{n-1} \\ & \downarrow & \downarrow \\ & E \otimes \mathbb{Z} & \xrightarrow{f} G_{n-1} \end{array}$$

By the induction hypothesis, $H^*(G_{n-1}) = \mathbb{Z}[c_1, \dots, c_{n-1}]$, so we get s.e.s.

$$0 \rightarrow H^i(G_n) \xrightarrow{\wedge} H^{i+2n}(G_n) \xrightarrow{\pi^*} H^{i+2n}(G_{n-1}) \rightarrow 0$$

Let $x \in H^{i+2n}(G_n)$. Then $\lambda x = p(c_1, \dots, c_{n-1})$ for a unique p . Then $x - p(c_1, \dots, c_{n-1}) = \gamma c_n$ for a unique γ . Inductively on i , $\gamma = q(c_1, \dots, c_n)$ for a unique q . Hence,

$$x = p(c_1, \dots, c_{n-1}) + q(c_1, \dots, c_n) c_n$$

uniquely.

Then: $c(w \otimes \varepsilon) = c(w) c(\varepsilon)$.

1. $c(w \otimes \varepsilon) = P(c(w), c(\varepsilon))$
2. $P = \text{product}$

Then: $c_i(w) = (-1)^i c_i(w)$

Inductively: For the Euler class, it is easy

Example (Chern class of $\mathbb{P}^n$): Let w^* be the orthogonal complement of γ^* , so $w \otimes \gamma^*$ is trivial. We will show that $\text{Hom}(\gamma^*, w^*) = \mathcal{T}$, the tangent bundle:

If $L \in \mathbb{P}^n$ then $\text{Hom}(L, L^\perp)$ is the set of graphs of linear maps $L \rightarrow L^\perp$. Restricting to the sphere, we get

$$\begin{array}{ccc} & L & \\ & \downarrow & \\ & L^\perp & \end{array}$$

We get

$$\begin{array}{ccc} \text{trivial} & & \\ \tau \otimes \varepsilon^* & = & \text{Hom}(\gamma^*, w^*) \oplus \text{Hom}(\gamma^*, \gamma^*) \\ & = & \text{Hom}(\gamma^*, w^* \oplus \gamma^*) \\ & = & \text{Hom}(\gamma^*, \mathbb{C}^{n+1}) = \bigoplus \gamma^* \end{array}$$

Note: This also gives the s.e.s.
 $D \rightarrow D_{\mathbb{P}^n} \rightarrow D_{\mathbb{P}^n}(1)^{n+1} \rightarrow \mathcal{T} \rightarrow 0$
since $\gamma^* = D(1)$.

Hence, $c(\mathcal{T}) = c(\tau \otimes \varepsilon^*) = \prod c(\gamma^*) = (1 - c_1)^{n+1}$.

Note: We have $n+1 = \chi(\mathbb{P}^n) = \int c_n(\mathcal{T}) = \int (n+1) c_1$.

Hence, $\int c_1 = -1$.

The Chern-Weil approach to characteristic classes

Let $\zeta: E \rightarrow B$ be a $\mathbb{C}^n$ -vector bundle, τ^* the cotangent bundle. Let

$$\nabla: \zeta \rightarrow \zeta \otimes \tau^*$$

be a connection. Note that locally, if $s_i: U \rightarrow E$ trivialise E ,

$$\nabla s_i = \sum_j w_{ij} \otimes s_j$$

for $w_{ij} \in \tau^*(U)$. We extend ∇ to a $\mathbb{C}$ -linear map $\zeta \otimes \tau^* \rightarrow \zeta \otimes \wedge^2 \tau^*$ by

$$\nabla(s \otimes \theta) = \nabla s \wedge \theta - s \otimes \partial \theta.$$

Let $K = \nabla^2$. Then K is a section of $\text{Hom}(\zeta, \zeta \otimes \wedge^2 \tau^*)$.

In a local trivialisation,

$$\begin{aligned} K(s_i) &= \nabla(\sum_j w_{ij} \otimes s_j) = \sum_j \nabla w_{ij} \otimes s_j - w_{ij} \wedge \sum_k w_{jk} \otimes s_k \\ &= \sum_{j,k} \Omega_{ik} \otimes s_k, \end{aligned}$$

where $\Omega_{ik} = \nabla w_{ik} - \sum_j w_{ij} \wedge w_{jk}$, or $\Omega = \partial w - w \wedge w$.

Let $P: M_n \rightarrow \mathbb{C}$ be an invariant polynomial, i.e. $P(TXT^{-1}) = P(X)$. We can apply $P(K)$ to get a differential form, by evaluating P on Ω in any local trivialisation (Note: Ω_{ij} are 2-forms, so the algebra they generate is commutative).

Prop: $\partial P(K) = 0$

Proof: We have $\partial P(K) = \frac{\partial P}{\partial \Omega_{ij}} \partial \Omega_{ij} = \text{Tr}(P'(\Omega) \partial \Omega)$.

Also $\partial \Omega = \partial(\partial w - w \wedge w) = \partial w \wedge \Omega - \Omega \wedge w$. Note that $P'(A)A = AP(A)$:

Differentiate $P((1+tE)A) = P(A(1+tE))$ and set $t=0$.

Hence, $\Omega \wedge P'(\Omega) = P'(\Omega) \wedge \Omega = X$, and

$$\begin{aligned} \partial P(K) &= \text{Tr}(P'(\Omega) \wedge w \wedge \Omega - P'(\Omega) \wedge \Omega \wedge w) \\ &= \text{Tr}(X \wedge \Omega - \Omega \wedge X) \\ &= X_{ij} \wedge \Omega_{ji} - \Omega_{ji} \wedge X_{ij} = 0 \end{aligned}$$

Note that the cohomology class of $P(K)$ is independent of ∇ , since we have a connection to $\nabla_1 + (1-t)\nabla_2$ on $B \times [0,1]$.

We now discuss connections on the cotangent bundle τ^* . We call such a connection symmetric if

$$\begin{array}{ccc} \tau^* & \xrightarrow{\nabla} & \tau^* \otimes \tau^* \\ \nabla \downarrow & \searrow & \uparrow \\ \tau^* \otimes \tau^* & & \end{array}$$

Note that ∇ is symmetric iff for any function, ∇f is symmetric:

We show this for $\partial \tau^*$. $\nabla \partial \tau^* = \sum_{ij} \nabla \partial^2 x^i \otimes \partial^2 x^j = \partial^2 x^k \otimes 0$ iff $\Gamma_{ij}^k = \Gamma_{ji}^k$.

If B is a Riemannian manifold then τ^* has a unique symmetric connection compatible with the metric, the Levi-Civita connection.

Thm (Gauss-Bonnet): Let ζ be a $\mathbb{C}$ -line bundle with a connection ∇ . Then

$$[K] = 2\pi i c_1(\zeta).$$

By naturality, $[K] \in H^2(B; \mathbb{C})$ is a characteristic class. Hence, $[K] = 2\pi i c_1(\zeta)$ for some $\zeta \in \mathbb{C}$, not depending on B or ζ (by Yano's lemma). We can find it by calculating the integral of both sides on e.g. τ^* of $\mathbb{P}^2 = \mathbb{S}^2$.

Note: The usual Gauss-Bonnet follows by giving the tangent bundle of a closed oriented Riemannian surface the complex structure coming from a rotation by $\frac{\pi}{2}$.

In general, if σ_r is the r -th symmetric polynomial, then

$$[\sigma_r(K)] = (2\pi i)^r c_r(\zeta).$$

This follows from the fact that for $\varepsilon(A) = \det(1 + \frac{A}{2\pi i}) = \sum \frac{\sigma_r(A)}{(2\pi i)^r}$, $\varepsilon(K) = c(\zeta)$ for line bundles, and from the splitting principle.

The splitting principle

Let $\pi: P(E) \rightarrow B$. We can split the tautological line bundle off $\pi^* E$ and continue with the quotient $\pi^* E / \text{ker } \mathcal{S}$. We arrive at a manifold $F(E) \xrightarrow{\pi} B$ s.t. $\pi^* E$ is a sum of line bundles. By induction, $H^*(B) \hookrightarrow H^*(F(E))$, so if a polynomial relation holds here, it holds on $H^*(B)$ too.

Example: Let $E = L_1 \oplus \dots \oplus L_n$. Then $\wedge^n E = L_1 \otimes \dots \otimes L_n$, so

$$c(\wedge^n E) = 1 + (c_1(L_1) + \dots + c_1(L_n)) + \text{h.o.t.}$$

In particular, $c_1(\wedge^n E) = c_1(E)$, which hence holds for any E .

We can describe the manifold $F(E)$ obtained above as the flag bundle $F(E) \rightarrow B$, which is obtained by replacing the fibres E_p by $F(E_p) = \text{Gr}_n / \text{Gr}_{n-1}$. To see why $F(E) = F(E)$, think it through for a rank 3 vector space:

$$\begin{array}{ccccc} V & \leftarrow & L_1 \otimes Q_1 & \leftarrow & L_1 \otimes L_2 \otimes Q_2 \\ \downarrow & & \downarrow & & \downarrow \\ P & \leftarrow & P(V) & \leftarrow & P(Q_1) \end{array}$$

We now calculate the cohomology of $P(E)$, and then, iteratively, the one of $F(E)$.

Note that the tautological line subbundle L of $\pi^* E$ restricts in $\pi^{-1}(p)$ to the taut. bundle on $P(E_p)$. Hence, its first Chern class x restricts to a generator of the cohomology of the fibres. Hence, by the Leray-Hirsch theorem,

$$H^*(P(E)) = H^*(B)[x] / x^n - \alpha_1 x^{n-1} + \dots + \alpha_n$$

for some $\alpha_i \in H^{2i}(B)$. We claim that $\alpha_i = c_i(E)$:

We know that $H^*(P(E))$ is generated by x . Let $y_i = c_i(E / \text{ker } \mathcal{S})$. Then

$$H^*(P(E)) = H^*(B)[x, y_1, \dots, y_{n-1}] / c(L) c(E / \text{ker } \mathcal{S}) = c(\pi^* E), \text{ the relation (identity)}$$

by the product formula. The relation gives the equation $(x + y_1 + \dots + y_{n-1})^n = 4c_1 x^{n-1} + \dots + c_n$, hence

$$\begin{array}{ccc} x + y_1 = c_1 & \rightsquigarrow & y_1 = c_1 - x \\ x y_1 + y_2 = c_2 & \rightsquigarrow & y_2 = c_2 - x y_1 = c_2 - x c_1 + x^2 \\ \vdots & & \\ x y_{n-1} = c_n & & \end{array}$$

Expressing each y_i in terms of x , we get that the relation is equivalent to the claim. In particular, there are no other relations.

This gives another definition of Chern classes. From the proof, it follows that

$$H^*(P(E)) = H^*(B)[c(L), c(E / \text{ker } \mathcal{S})] / c(L) c(E / \text{ker } \mathcal{S}) + c(\pi^* E).$$

This can be used to prove inductively that $x_i = c_i$

$$H^*(F(E)) = H^*(B)[x_1, \dots, x_n] / \prod (1 + x_i) = c(\pi^* E).$$

We have

$$\begin{array}{ccccccc} E & \leftarrow & L \otimes Q_1 & \leftarrow & L \otimes L_2 \otimes Q_2 & \leftarrow & \dots \leftarrow L \otimes \dots \otimes L_{n-1} \otimes Q_n \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ B & \leftarrow & P(E) & \leftarrow & P(Q_1) & \leftarrow & \dots \leftarrow F(E). \end{array}$$

Then

$$\begin{aligned} H^*(P(Q_1)) &= H^*(P(E)) [c(L_2), c(Q_2)] / c(L_2) c(Q_2) = c(Q_1) \\ &= H^*(B) [c(L_1), c(L_2), c(Q_1), c(Q_2)] / c(L_1) c(L_2) c(Q_1) c(Q_2) \\ &= H^*(B) [c(L_1), c(L_2), c(Q_2)] / c(L_1) c(L_2) c(Q_2) = c(E). \end{aligned}$$